

Numerical Analysis PhD Exam, May 2018, Part 2

Do all five (5) problems.

1. Consider

$$y' = 4ty, \quad t \in [0, 1], \quad y(0) = 2, \quad (1)$$

which has solution $Y(t) = 2e^{2t^2}$.

- (a) Using the Euler method error estimate find the value of h required to ensure that the Euler method solution w_i of (1) satisfies $|Y(t_i) - w_i| \leq 0.1$ for all i .
- (b) Derive the Taylor method of order 2 for (1).

2. Let $g \in C^2([a, b])$ and $p \in (a, b)$ with $g(p) = p$, $g'(p) = 0$, $g''(p) \neq 0$.

- (a) Show there is an $\epsilon > 0$ so that for all $x \in [p - \epsilon, p + \epsilon]$, we have $g^n(x) \rightarrow p$ as $n \rightarrow \infty$.
- (b) With ϵ as in part (a), show that for all $x \in [p - \epsilon, p + \epsilon]$, we have $|g(x) - p| \leq M|x - p|^2$, where $M = \max\{|g''(x)| : |x - p| \leq \epsilon\}/2$.

3. Find the second order (i.e. quadratic) least squares approximation to the function $f(x) = x^4$ on the interval $[-1, 1]$ with respect to the weight $w(x) \equiv 1$ using the fact that the first three orthonormal polynomials on $[-1, 1]$ with respect to this weight are

$$\varphi_0(x) = \sqrt{1/2}, \quad \varphi_1(x) = \sqrt{3/2}x, \quad \text{and} \quad \varphi_2(x) = \sqrt{5/8}(3x^2 - 1).$$

4. This problem has two parts.

- (a) Assume $N(h)$ is the computed approximation for M for each $h > 0$ and

$$M = N(h) + C_1h + C_2h^2 + C_3h^3 + \dots$$

Use the values $N(h)$, $N(h/3)$, and $N(h/9)$ to produce a $O(h^3)$ approximation to M .

- (b) Taylor's formula yields the following.

$$f'(x_0) = \frac{1}{h} \left(f(x_0 + h) - f(x_0) \right) - \frac{h}{2} f''(x_0) - \frac{h^2}{6} f'''(x_0) + O(h^3).$$

Use this with the extrapolation of part (a) to derive an $O(h^3)$ formula for $f'(x_0)$.

5. Find a, b, c so that the quadrature formula

$$\int_0^2 f(x) dx = af(0) + bf(1) + cf(2)$$

has degree of precision as large as possible, and show it is no larger than your answer.